

Waves

⇒ any physical quantity that varies with space time as $y = f(x \pm vt)$

$v = \text{const}$

$t = \text{time}$

$\phi(x, y, z)$

$$d\phi = \frac{d\phi}{dx} dx + \frac{d\phi}{dy} dy + \frac{d\phi}{dz} dz$$

$$\Rightarrow \left(\hat{x} \frac{d\phi}{dx} + \hat{y} \frac{d\phi}{dy} + \hat{z} \frac{d\phi}{dz} \right) \cdot (\hat{x} dx + \hat{y} dy + \hat{z} dz)$$

Let $\vec{r} = \hat{x}x + \hat{y}y + \hat{z}z$

$d\vec{r} = \hat{x} dx + \hat{y} dy + \hat{z} dz$

$$\Rightarrow (\vec{\nabla} \phi) \cdot \vec{n} d\vec{r}$$

$$\frac{d\phi}{dr} \Rightarrow (\vec{\nabla} \phi) \cdot \hat{n} \rightarrow \text{directional derivative}$$

⇒ gradient for a level curve is always normal to the surface

→ In one dimension

$$\vec{E} = -\frac{d\phi}{dn} \hat{n}$$

for 3D

$$\vec{E} = -\left(\frac{d\phi}{dn} \hat{n} + \frac{d\phi}{dy} \hat{y} + \frac{d\phi}{dz} \hat{z}\right) = -\frac{d\phi}{dn}$$

→ gravitational and electric fields are conservative fields here work done in moving mass along closed loop is zero and potential diff b/w 2 points doesn't depend on mass/charge.

flux: $\int \vec{V} \cdot d\vec{s}$ open surface

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Total flux through cuboid.

$$\oint_{\text{closed}} \vec{V} \cdot d\vec{s} = \lim_{dv \rightarrow 0} \left(\frac{\delta V_x}{\delta x} + \frac{\delta V_y}{\delta y} + \frac{\delta V_z}{\delta z} \right) dv$$

divergence $\Rightarrow$ Total outward flux by small volume

we can divide the total volume into tiny boxes and add total outward flux through all tiny boxes

$$\oint_{S_{\text{big}}} \vec{V} \cdot d\vec{s} = \oint_{V_{\text{big}}} (\vec{\nabla} \cdot \vec{V}) dv$$

$$\Rightarrow \lim_{\substack{\text{all tiny} \\ \text{vol}}} \oint_{V} (\vec{\nabla} \cdot \vec{V}) dv \Rightarrow \oint \vec{V} \cdot d\vec{s}$$

$$\Rightarrow \oint_{\substack{\text{unshared} \\ \text{faces coinciding with} \\ \text{of big volume}}} \vec{V} \cdot d\vec{s} \Rightarrow \oint_{\substack{\text{entire surface} \\ \text{boundary}}} \vec{V} \cdot d\vec{s}$$

Gauss's Divergence Theorem

$$\int_V (\vec{\nabla} \cdot \vec{V}) dv = \oint_S \vec{V} \cdot d\vec{s}$$

$$\rightarrow \vec{\nabla} \cdot (\vec{\nabla} V) \Rightarrow \left(\hat{x} \frac{d}{dx} + \hat{y} \frac{d}{dy} + \hat{z} \frac{d}{dz} \right) \cdot \left(\hat{x} \frac{dV}{dx} + \hat{y} \frac{dV}{dy} + \hat{z} \frac{dV}{dz} \right)$$

Div of grad of V

$$\Rightarrow \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = \nabla^2 V$$

$$\star \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \rightarrow \text{laplacian operator}$$

Earnshaw's Theorem

$\star$ A scalar field $\phi(x, y)$ that has $\nabla \phi = 0$ over a region in space cannot have local maxima or minima in that region. we can at most have saddle point

Curl operator

$$\vec{\nabla} \times A = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{d}{dx} & \frac{d}{dy} & \frac{d}{dz} \\ A_x & A_y & A_z \end{vmatrix}$$

$$\Rightarrow \left(\frac{dA_z}{dy} - \frac{dA_y}{dz} \right) \hat{x} - \left(\frac{dA_z}{dx} - \frac{dA_x}{dz} \right) \hat{y} + \left(\frac{dA_y}{dx} - \frac{dA_x}{dy} \right) \hat{z}$$

Curl as line integral

$\vec{s}$ circulation

$$\oint \vec{A} \cdot d\vec{L} = \left\{ A_y(x_0 y_0 z_0) - \frac{dz}{2} \frac{\partial A_y}{\partial z} \right\} dy$$

$$- \left\{ A_y(x_0 y_0 z_0) + \frac{dz}{2} \frac{\partial A_y}{\partial z} \right\} dy + \left\{ A_z(x_0 y_0 z_0) + \frac{dy}{2} \frac{\partial A_z}{\partial y} \right\} dz - \left\{ A_z(x_0 y_0 z_0) - \frac{dy}{2} \frac{\partial A_z}{\partial y} \right\} dz$$

$$\Rightarrow \oint \vec{A} \cdot d\vec{L} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) dy dz$$

$$\Rightarrow \text{curl} = \lim_{dS \rightarrow 0} \frac{\oint \vec{A} \cdot d\vec{L}}{dS}$$

Stokes thm,

$$\int (\vec{\nabla} \times \vec{A}) \cdot d\vec{S} = \oint \vec{A} \cdot d\vec{L}$$

if curl of field is 0 it is irrotational field
irrotational field is conservative field

* gradient = ∇A

Divergence = $\nabla \cdot A \rightarrow$ scalar

Curl = $\nabla \times A$

Laplacian = $\nabla^2 A$

Solenoidal field

$$\vec{\nabla} \cdot \vec{F} = 0 \quad \text{Solenoidal field} \rightarrow \text{divergence}$$

→ Magnetic field is solenoidal field since it has no source or sink.

→ Electrostatic field is solenoidal only if no charge is present

Tut

1) $\vec{f} = 3x^2\hat{i} + 5xy^2\hat{j} + xyz^3\hat{k}$

$$6x + 10xy + 3xyz^2$$

$$\Rightarrow 6 + 20 + 54 \Rightarrow \underline{80}$$

2) $\vec{D} = r \sin \phi \hat{a}_r - \frac{1}{r} \sin \theta \cos \phi \hat{a}_\theta + r^2 \hat{a}_\phi$

$$\vec{D} = \vec{D}_{\text{rad}} + \vec{D}_T$$

$$\Rightarrow \vec{D}_{\text{rad}} - \vec{D}_{\text{no}} = \vec{D}_T = -0.048\hat{\theta} + 100\hat{\phi}$$

$\theta = \text{polar}$

$\phi = \text{azimuth}$

c)

r	θ	ϕ
-5	-0.043	100
0	1	0

$$\Rightarrow 5\hat{\theta} - 100\hat{r}$$

is vector along cone

$$\vec{A} = 3\hat{a}_x + 2\hat{a}_y - 6\hat{a}_z$$

$$\vec{B} = 4\hat{a}_x + 3\hat{a}_y$$

$$a) -6$$

$$b) \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ 3 & 2 & -6 \\ 4 & 0 & 3 \end{vmatrix} = 6\hat{a}_x - (8\hat{a}_y - 12\hat{a}_z) = \sqrt{6^2 + 3^2 + 8^2} = 34.48$$

c) component along $\hat{a}_z \Rightarrow$

$$(\vec{A} \cdot \hat{a}_z) \hat{a}_z = \hat{a}_z = \cos\theta \hat{a}_x - \sin\theta \hat{a}_y$$

$$\Rightarrow \frac{\cos\pi}{3} \hat{a}_x - \frac{\sin\pi}{3} \hat{a}_y$$

$$\Rightarrow \frac{1}{2} \hat{a}_x - \frac{\sqrt{3}}{2} \hat{a}_y$$

$$\left(\frac{3}{2} - \sqrt{3}\right) \hat{a}_z \Rightarrow \left(\frac{3}{2} - \sqrt{3}\right) \left(\frac{1}{2} \hat{a}_x - \frac{\sqrt{3}}{2} \hat{a}_y\right)$$

$$T = \ln(1/\rho) \quad | \quad \vec{F} = -\vec{\nabla} T$$

$$\Rightarrow -\frac{\partial}{\partial \rho} \ln\left(\frac{1}{\rho}\right) \hat{a}_\rho$$

$$\Rightarrow \frac{1}{\rho} \hat{a}_\rho$$

$$\vec{B} \cdot \vec{F} = \frac{1}{\rho} \frac{\partial}{\partial \rho} \ln \rho \Rightarrow \frac{1}{\rho^2}$$

$$\text{flux} \Rightarrow \oint \vec{F} \cdot \hat{n} dS \Rightarrow$$

~~Q~~ Divergence

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla \cdot \mathbf{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

spherical $\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$

- Vector field is solenoidal if $\nabla \cdot \mathbf{A} = 0$
- Vector field $\mathbf{A}$ is irrotational if $\nabla \times \mathbf{A} = 0$ ↖ curl

Vector identities

i) $\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$

ii) $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$ | iii) $\vec{\nabla} \times (\vec{\nabla} \cdot \vec{U}) = 0$

iv) $\vec{\nabla} \cdot (\vec{\nabla} U) = \nabla^2 U \rightarrow \text{laplacian}$

v) $\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A}$

vi) $\vec{\nabla} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B})$

dirac delta fn

$$\delta(x-x_0) = \begin{cases} 0 & x \neq x_0 \\ \infty & x = x_0 \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(x-x_0) dx = 1$$

$$\star \int_{-\infty}^{\infty} f(x) \delta(x-x_0) dx = f(x_0)$$

$$\star \delta(kx) = \frac{1}{|k|} \delta(x)$$

dirac delta fn in 3D

$$\rightarrow \delta^3(\vec{r} - \vec{r}_0) = \delta(x-x_0) \delta(y-y_0) \delta(z-z_0)$$

$$\iiint \delta^3(\vec{r} - \vec{r}_0) = \int \delta(x-x_0) \int \delta(y-y_0) \int \delta(z-z_0) = 1 //$$

1) if $\vec{F} = \frac{\hat{r}}{r^2}$ not diff at $r=0$
 $\nabla \cdot \vec{F} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(\frac{r^2}{r^2} \right) = \frac{1}{r^2} \times 0 = 0$ // ∇
 $\nabla \cdot \vec{F} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(\frac{r^2}{r^2} \right) = \frac{1}{r^2} \times 0 = 0$ // ∇

↳ by Gauss div thm

$\int_{-\infty}^{\infty} (\nabla \cdot \vec{F}) dV = \oint \vec{F} \cdot d\vec{s} = \oint \frac{\hat{r}}{r^2} r^2 \sin \theta d\theta d\phi$
 $\Rightarrow \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta$ lim fixed for spherical
 $2\pi (\cos \theta) \Big|_0^\pi \Rightarrow 2\pi (1 - (-1)) = 4\pi$

hence

$\int (\nabla \cdot \vec{F}) dV = 0, r \neq 0$
 $(4\pi, r=0)$

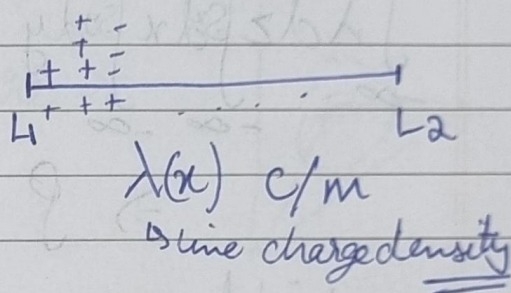
$\therefore (\nabla \cdot \vec{F}) = 4\pi \delta^3(r)$
 $\Rightarrow \vec{\nabla} \cdot \frac{\hat{r}}{r^2} = 4\pi \delta^3(r)$

so similarly
 $\vec{\nabla} \cdot \left(\frac{\hat{r}}{(r-r_0)^2} \right) = 4\pi \delta^3(r-r_0)$

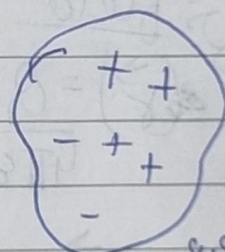
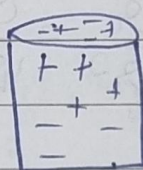
$$\begin{aligned}\nabla^2 \left(\frac{1}{r} \right) &\Rightarrow \nabla \cdot \left(\nabla \frac{1}{r} \right) = \nabla \cdot \left(-\frac{\hat{r}}{r^2} \right) \\ &\Rightarrow -\nabla \cdot \frac{\hat{r}}{r^2} \\ &\Rightarrow -4\pi \delta^3(r) //\end{aligned}$$

Electrostatics

i) $Q_{\text{total}} = \int_{L_1}^{L_2} \lambda(x) dx$



ii) $Q_{\text{total}} = \int_S \sigma(x, y) dA$



iii) $Q_{\text{total}} = \int_V \rho(x, y, z) dV$

$\rho(x, y, z)$
 $\rho(P, \theta, \phi)$
 $\rho(P, \phi, z)$

$\sigma(x, y)$
surface charge density

for point charge since $dV \rightarrow 0$

ρ has to $\rightarrow \infty$ since
 ρdV has to be finite

→ calculating vol charge density for line charge.

wkt $Q_{\text{total}} = \int_{-\infty}^{\infty} \lambda dz$ which is equal to

$$\int_{-\infty}^{\infty} \lambda dz \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy \Rightarrow \int_{-\infty}^{\infty} \lambda dz \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx dy \quad \leftarrow \int \rho dx dy dz \text{ now we find } \rho$$

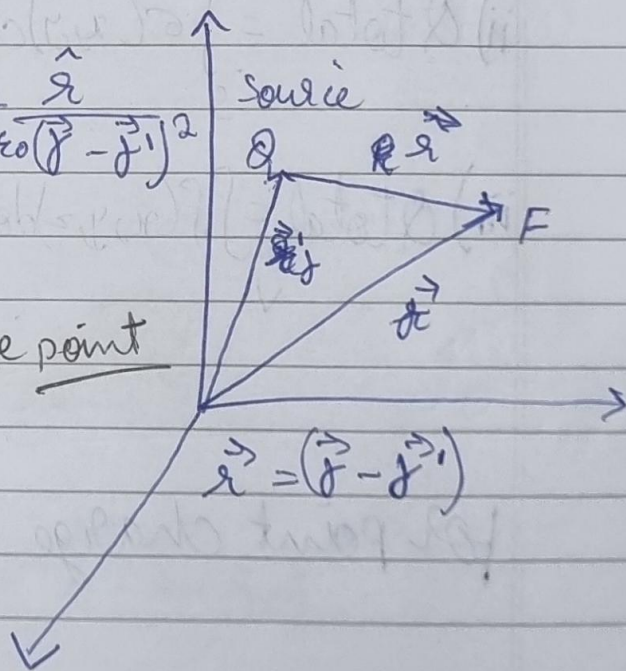
$$\Rightarrow \rho = \lambda \delta(x) \delta(y)$$

field

$$\vec{E}(\vec{r}) = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r} = \frac{Q}{4\pi\epsilon_0 (\vec{r} - \vec{r}')^2}$$

$$\vec{E} = -\nabla \phi$$

varying source point



Lets assume $E = 1/r^2$
this can be written as

$$\vec{E} = \frac{1}{r^2} = -\vec{\nabla} \left(\frac{1}{r} \right)$$

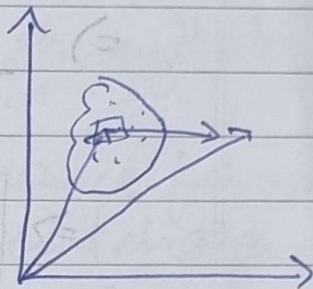
$$r = \vec{r} - \vec{r}'$$

$$\text{ie } \phi = -\frac{Q}{4\pi\epsilon_0} \vec{\nabla} \left(\frac{1}{r} \right)$$

we can do this
because of chain rule

The Electric field here $\Rightarrow$

$$\vec{E}(\vec{r}) = \int \frac{Q \hat{r}}{4\pi\epsilon_0 r^2}$$



we can write $Q = \int (x'y'z') \underbrace{dx'dy'dz'}_{dv'}$ we will take ρ

$$\Rightarrow \vec{E} = \int \frac{\rho(x'y'z')}{4\pi\epsilon_0 r^2} dx'dy'dz'$$

$$\vec{\nabla} \cdot \left(\frac{\hat{r}}{r^2} \right) = 4\pi \delta^3(\vec{r}) \Rightarrow 4\pi \delta^3(\vec{r} - \vec{r}')$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int \rho(\vec{r}') dv' \left(\frac{\hat{r}}{r^2} \right) \quad (a)$$

$$\Rightarrow \vec{E} = -\vec{\nabla} \phi \Rightarrow \phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho dv'}{r}$$

from (a)

$$\vec{\nabla} \cdot \vec{E} = \frac{1}{4\pi\epsilon_0} \int_V \vec{\nabla} \cdot \left(\rho(\vec{r}') \frac{\hat{r}}{r^2} \right) dV$$

$$\Rightarrow \frac{1}{4\pi\epsilon_0} \int_V \rho(\vec{r}') dV' \left(\vec{\nabla} \cdot \left(\frac{\hat{r}}{r^2} \right) \right)$$

$$\Rightarrow \frac{1}{\epsilon_0} \int_V \rho(\vec{r}') dV' \delta^3(\vec{r}' - \vec{r}) \quad \rightarrow \text{due to symmetric fn}$$

$$\Rightarrow \boxed{\frac{\rho(\vec{r})}{\epsilon_0} = \vec{\nabla} \cdot \vec{E}}$$

$\rightarrow$ Gauss's Law (differential form) $\checkmark$

$$\oint \vec{E} \cdot d\vec{s} = \frac{Q_{enc}}{\epsilon_0}$$

$\rightarrow$ closed surface

$$\iiint (\vec{\nabla} \cdot \vec{E}) dV = \frac{1}{\epsilon_0} \int \rho dV = \frac{Q_{enc}}{\epsilon_0}$$

curl of $\vec{E}$ is 0 since it is a conservative field.

curl of gradient is 0

wkt $\phi = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$

$$\vec{E} = -\vec{\nabla} \phi$$

$$\vec{\nabla} \times \vec{E} = \vec{\nabla} \times (-\vec{\nabla} \phi) \Rightarrow \text{for point charge} \Rightarrow 0$$

wkt $\vec{E} = \sum_i \vec{E}_i$ for multiple point

$$\Rightarrow \vec{\nabla} \times \vec{E} = \sum_i \vec{\nabla} \times \vec{E}_i = 0 \text{ hence for non point charge also}$$

Only Electrostatic field is conservative

Potential difference

Q_{BA} = charge taken from A to B

$$W = \int_A^B -\vec{E} \cdot d\vec{l} \Rightarrow - \int_A^B (\vec{\nabla} \phi) \cdot d\vec{l}$$

$$\Rightarrow \boxed{\int_A^B d\phi \Rightarrow \phi(B) - \phi(A) //$$

mag field

$$\vec{B} = \vec{\nabla} \times \vec{A} \rightarrow \text{corresponding potential}$$

$$\oint \vec{B} \cdot d\vec{s} = 0$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

hence here we will define

$$\vec{B} \Rightarrow (\vec{\nabla} \times \vec{A})$$

↳ curl of A (pot)

here pot is vector since we have to represent it as curl of something

$$\vec{A}_{\text{new}} = \vec{A} \pm (\vec{\nabla} \times)$$

we take an $\vec{A}_{\text{new}}$ such that

curl of $\vec{A}_{\text{new}}$ always comes out to $\vec{A}$ hence we add a gradient of $\times$ since curl of grad is always 0 //

We do this since it gives us opportunity to take div as anything //

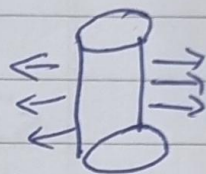
$$\phi(r) = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda(r')}{r} dr'$$

$$\phi(r) = \frac{1}{4\pi\epsilon_0} \iint \frac{\sigma(r')}{r} d\Omega'$$

$$\phi(r) = \frac{1}{4\pi\epsilon_0} \iiint \frac{\rho(r')}{r} dv'$$

Q) $\vec{G}(r) = 10e^{-2z}(\rho \hat{a}_\rho + \hat{a}_z)$

$\rho = 1 \mid 1 \geq z \geq 0 \mid, \phi = 0, 2\pi$



Wkt $\Psi_{\text{total}} = \Psi_{\text{side}} + \Psi_{\text{bottom}} + \Psi_{\text{top}}$

$$\int \vec{G} \cdot d\vec{s} = \Psi$$

$$\Psi = \int \vec{G} \cdot d\vec{s}$$

$$\Rightarrow \int 10e^{-2z} (\rho \hat{a}_\rho) \times \rho dz d\phi \hat{a}_\rho$$

$$\Rightarrow \int 10e^{-2z} \rho^2 dz d\phi$$

$$\Rightarrow \int_0^1 \int_0^{2\pi} 10e^{-2z} dz d\phi$$

$$\Rightarrow 10 \int_0^1 e^{-2z} dz \int_0^{2\pi} d\phi$$

$$ds_s = \rho dz d\phi \hat{a}_\rho$$

$$ds_b = \rho d\phi d\hat{a}_z$$

$$ds_T = \rho d\rho d\hat{a}_z$$

$$\Psi = 10\pi(1-e^2)$$

$$\Rightarrow 10 \left[\frac{e^{-2}}{-2} - 1 \right] \times 2\pi \Big|_0^1$$

Using Divergence Theorem

$$\Psi = \oint_S \vec{G} \cdot d\vec{s} = \int_V (\vec{\nabla} \cdot \vec{G}) dV$$

$$\vec{\nabla} \cdot \vec{G} = \frac{1}{\rho} \frac{\partial}{\partial \rho} \rho G_\rho + \frac{1}{\rho} \frac{\partial}{\partial \phi} G_\phi + \frac{\partial}{\partial z} G_z$$

$$\Rightarrow \frac{1}{\rho} \frac{\partial}{\partial \rho} \rho^2 \times 10 e^{-2z} + \frac{1}{\rho} \frac{\partial}{\partial \phi} 0 + \frac{\partial}{\partial z} 10 e^{-2z}$$

$$\Rightarrow \frac{1}{\rho} \times 2\rho \times 10 e^{-2z} + 0 - 20 e^{-2z}$$

$$\Rightarrow 20 e^{-2z}$$

(?)

$$\Psi_b = \int_b \vec{G} \cdot d\vec{s} = \Phi$$

$$\Rightarrow \int_0^1 \int_0^{2\pi} 10e^{-2z} \hat{a}_z \rho d\rho d\hat{\phi} \hat{a}_z$$

$$\Rightarrow \int_0^1 \int_0^{2\pi} 10e^{-2z} \rho d\rho d\hat{\phi} \Rightarrow 2\pi \times 10e^{-2 \times 0} \times \frac{1}{2} \Rightarrow 20\pi \times \frac{1}{2} = 10\pi$$

$$\Psi_t = \int_T \vec{G} \cdot d\vec{s}$$

$$\Rightarrow \int_0^1 \int_0^{2\pi} 10e^{-2z} \hat{a}_z \rho d\rho d\hat{\phi} \hat{a}_z$$

$$\Rightarrow 10e^{-2 \times 1} \times \frac{1}{2} \times 2\pi \Rightarrow 10\pi e^{-2}$$

$$\Psi = \Psi_t + \Psi_b + \Psi_{\text{side}}$$

Q2) a) $\vec{A} = x^2 y \hat{a}_x + y^2 z \hat{a}_y - 2xz \hat{a}_z$

$\vec{\nabla} \times \vec{A}, \vec{\nabla} \cdot (\vec{\nabla} \times \vec{A})$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{d}{dx} & \frac{d}{dy} & \frac{d}{dz} \\ x^2 y & y^2 z & -2xz \end{vmatrix}$$

$$(0 - y^2) \hat{x} - \hat{y}(-2z - 0) + \hat{z}(0 - 2x^2)$$

$$\Rightarrow -y^2 \hat{x} + 2z \hat{y} - x^2 \hat{z}$$

$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$

b) $\begin{vmatrix} \hat{\rho} & \hat{\phi} & \hat{z} \\ \frac{d}{d\rho} & \frac{d}{d\phi} & \frac{d}{dz} \\ \rho^2 & \rho^3 & 3\rho z^2 \end{vmatrix}$

$$0 \hat{\rho} - \hat{\rho}(3z^2 - 0) + \hat{z}(\rho^2 - 0)$$

$$\Rightarrow (\rho^2 - 3z^2) \hat{\phi} + 3\rho^2 \hat{z}$$

$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A})$

Q3) Let scalar field $= V(x, y, z)$

show that

$$\vec{\nabla} \times \vec{\nabla} V = 0$$

$$\vec{\nabla} V \Rightarrow \frac{d}{dx} V(x, y, z) \hat{i} + \frac{d}{dy} V(x, y, z) \hat{j} + \frac{d}{dz} V(x, y, z) \hat{k}$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{d}{dx} & \frac{d}{dy} & \frac{d}{dz} \\ \frac{dV(x,y,z)}{dx} & \frac{dV(x,y,z)}{dy} & \frac{dV(x,y,z)}{dz} \end{vmatrix}$$

$$\Rightarrow \left(\frac{d^2}{dy dz} (V(x,y,z)) - \frac{d^2}{dz dy} (V(x,y,z)) \right) \hat{i} + 0 \hat{j} + 0 \hat{k}$$

↑
equal $\Rightarrow 0$

$$\therefore 0 //$$

Q4) a) $V = x^3 y^2 e^{xz}$ $P = (1, -1, 1)$ (8)

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\frac{\partial}{\partial x} (3x^2 y^2 e^{xz} + x^3 y^2 e^{xz}) + \frac{\partial}{\partial y} (2y x^3 e^{xz})$$

$$+ \frac{\partial}{\partial z} (x^3 y^2 e^{xz} \cdot xz)$$

$$\Rightarrow 6xy^2 e^{xz} + 2x^3 y^2 e^{xz} + x^3 y^2 z^2 e^{xz} + 2x^3 y^2 e^{xz} + x^5 y^2 e^{xz}$$

$$\Rightarrow 43.49$$

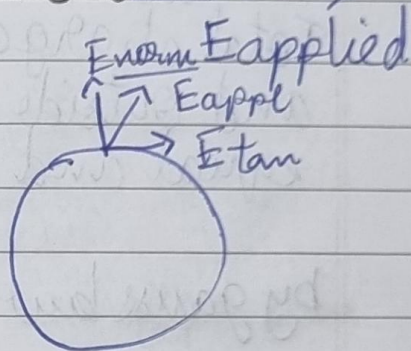
b) $\frac{1}{p}$

$$\& E_{\text{total}} = E_{\text{applied}} + E_{\text{induced}}$$

time taken for induced E_{field} is instantaneous

★ hence total electric conductor (PEC) (Perfect) is always $\emptyset$

The tangential E_{field} is $\emptyset$ since if it was not $\emptyset$ field would be induced and it would make it $\emptyset$ until field imbalance goes away.

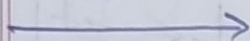


- may not be
- i) The E_{normal} is ~~not~~ equal to $\emptyset$
 - ii) If E_{normal} is pointed inside it is always $\emptyset$ since it is inside conductor.

Since E_{tan} is $\emptyset$ potential of adjacent points is equal and hence surface forms eq potential surface.

I.e surface of PEC is equipotential.

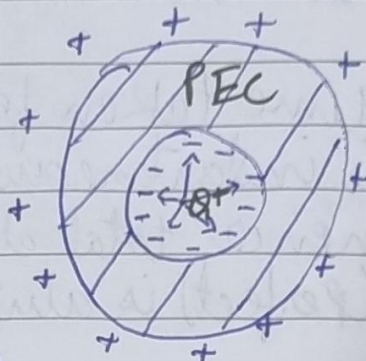
4) $\vec{\nabla} \cdot \vec{E} = 0$ inside conductor i.e. $\rho = 0$



→ Inside PEC field

is still 0. ^{induced}

Total charge on inside and outside surface is equal and equal to Q (-ve and +ve)

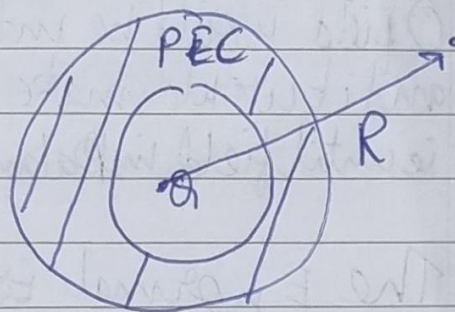


by gauss law

$$\oint \vec{E} \cdot d\vec{s} = \frac{Q_{enc}}{\epsilon_0}$$

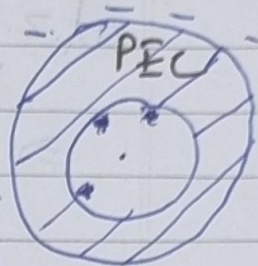
$$\Rightarrow E \times 4\pi R^2 = \frac{Q}{\epsilon_0}$$

$$\Rightarrow E = \frac{Q}{4\pi R^2 \epsilon_0}$$



If we are close direction of field might not be perpendicular but it is always applicable if point is far

why → The field inside the PEC is always going to be 0



On outside surface charge will be distributed such that E tangential will be 0

Gauss's Law

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} = \vec{\nabla} \cdot (-\vec{\nabla} \phi) = -\vec{\nabla}^2 \phi$$

$\vec{E}$
Laplacian

$$\boxed{\frac{\rho}{\epsilon_0} = -\vec{\nabla}^2 \phi} \rightarrow \text{Poisson's eqn}$$

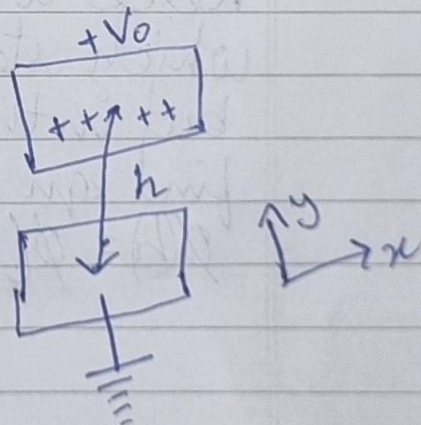
if $\boxed{\rho = 0} \Rightarrow \boxed{-\vec{\nabla}^2 \phi = 0} \rightarrow \text{Laplace eqn}$

by Lap

wkt $\frac{d^2 \phi}{dz^2} = 0$ (a)

$\Rightarrow \frac{d\phi}{dz} = c_1$ (b)

$\Rightarrow \phi = c_1 z + c_2$ (c)



$$\text{at } z=0$$

$$V_0 = C_2$$

$$\Rightarrow \phi = C_1 z + V_0$$

$$\Rightarrow \phi = C_1 h + V_0 = 0$$

$$\Rightarrow C_1 = -\frac{V_0}{h}$$

hence

$$\phi = -\frac{V_0}{h} z + V_0 \rightarrow \rho$$

② ϕ is a function only consisting of z terms since it is infinitely big in x and y directions hence change become 0.

③ Infinite Ladder

here we do not know which step we are in but in the step we can find eqn for ϕ which will be $\phi(h) = \phi(h+a)$

